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Avian Pathogenic *Escherichia coli* Isolated in Poultry Farms in Bangladesh that Use Antibiotics Extensively

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Colibacillosis caused by avian pathogenic *Escherichia coli* (APEC) is causing economic losses to the global poultry industry. Increased prevalence of antibiotic resistance in APEC is the leading cause for increased indiscriminate use of various antimicrobial compounds in farms. The study aimed to investigate the presence of phenotypic and genotypic markers for antibiotic resistance, metals, and biocides in APEC from Bangladeshi poultry and details about the antimicrobials used in poultry farms. In total, 55 APEC were isolated from hearts or liver samples of 86 sick or dead chickens using culture on agar plate and biochemical testing. APEC isolates were tested for antibiotic susceptibility to 14 antimicrobial agents according to the European Committee on Antimicrobial Susceptibility Testing. A series of PCRs was performed to screen the presence of genes for quinolones, colistin, aminoglycosides, ESBL, metals, and biocides. Detailed information regarding antibiotic use was collected from farmers during clinical investigations. Resistance was found to 10 antibiotics and prevalence was as follows: ampicillin (86%), ciprofloxacin (86%), trimethoprim-sulphamethoxazole (73%), chloramphenicol (33%), mecillinam (13%), gentamicin (11%), cefoxitin (11%), cefotaxime (9%), tigecycline (2%), and nitrofurantoin (2%). The most common multiresistance phenotype was CIP-AMP-SXT, and 35% of isolates were multidrug resistant. Genotypic analysis confirmed the presence of quinolone resistance genes [*qnrS1* and *aac-(6')-lb-cr*], silver-resistant genes (*silE*), and mercury-resistant genes (*merA*) but not others. In total, 88% farmers were using different antimicrobial compounds, and, of them, 56% were using antimicrobials without prescriptions from veterinarians. Ciprofloxacin was most extensively used followed by oxytetracycline. Critically important antibiotics like ciprofloxacin, colistin, and gentamicin are extensively used in the farms. This study confirmed the presence of antibiotics, metals, and biocide-resistant APEC in poultry farms in Bangladesh. Increased resistance to quinolones is a serious ongoing problem. The indiscriminate use of antibiotics in poultry farms is alarming and should be stopped.

Keywords: antibiotic resistance, antibiotic use APEC, poultry, farms, Bangladesh

Introduction

Escherichia coli is considered normal gut flora of humans, animals, and poultry, but some of the *E. coli* strains could be pathogenic to livestock and animals. Avian pathogenic *E. coli* (APEC) is considered a pathogen in poultry causing the disease called colibacillosis.¹ The pathogenic potential of APEC is high and it can invade different organs of birds and cause pericarditis, air-sacculitis, perihepatitis, and peritonitis.² APEC infection is generally found in chickens between 4 and

6 weeks of age and is considered a principal cause of morbidity and mortality, leading to considerable economic losses to the global poultry industry.¹

The rapid spread of antibiotic resistance is an emerging threat globally for both human and livestock populations and is considered a One Health problem.³ Indiscriminate and over-use of various antimicrobial compounds in human and veterinary medicine is one of the major forces behind the emergence of antimicrobial resistance in bacteria. Resistance to broad-spectrum antimicrobials reserved for human use has

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now become common in veterinary pathogens as these antimicrobial compounds are frequently used in livestock and aquaculture production.^{4–6} Various species of bacterial pathogens of pet animals and livestock are becoming more and more resistant to the latest generations of broad-spectrum antibiotics belonging to different classes, which ultimately cause infections that are difficult to treat or deaths in animals.⁷ Like other bacterial pathogens in poultry, antibiotic resistance in APEC is a serious challenge for the poultry industry. Several studies indicated an increased prevalence of antibiotic resistance and the magnitude of multidrug resistance among APEC isolates from different poultry species, more specifically in commercial poultry production.^{4,7,8}

Heavy metals and biocides (e.g., silver, copper, mercury) are used in agriculture, food storage, animal feed, farming, hospitals, and industrial benefits, now bacterial resistance to these compounds and their relationship with co-selection of antibiotic resistance is a major concern.^{9–11} Alarming, *E. coli* has started to show resistance to heavy metals (e.g., mercury) and biocide compounds (e.g., silver) because of the over- and misuse of these compounds.¹² There is no information available on poultry pathogens like APEC that could be potentially impacted by metal and biocide resistance biomarkers as biocide and metal compounds are used in feed and to ensure farm biosecurity.

Over the decades, poultry farming has been one of the most popular agro-industries in Bangladesh, and livestock and poultry farming contributes around 1.47% to the gross domestic product of Bangladesh.¹³ APEC is among the bacterial diseases of poultry in Bangladesh causing severe economic losses and increasing production costs through the use of different classes of broad-spectrum antibiotics.^{4,14} Antibiotic use in the poultry industry is indiscriminate and 80% of farmers were reported to use antibiotic-cocktail combinations in their farms without prescriptions from veterinarians.⁴ Multidrug resistance in APEC has appeared as a serious challenge in Bangladesh and APEC strains were reported to be resistant to antibiotics that are used in both veterinary and human medicine.^{4,14} There have been few studies describing the antibiotic resistance situation in APEC in Bangladesh and more studies are required as misuse of antimicrobial compounds increases day by day in the poultry sector. The study aimed to investigate antimicrobial susceptibility and genotypic markers for antibiotic resistance, metals, and biocides in APEC isolated from poultry in Bangladesh. Additionally, details about the pattern of antimicrobial compounds used by the farmers were investigated as there were many field poultry veterinarians who complained about the use of antibiotics by poultry farmers without veterinary prescriptions (personal communication of poultry vets in that region).

Materials and Methods

Sample collection

From June to October in 2016, 86 chickens suspected of having *E. coli* infection from 84 different farms (backyard, small-scale, and large-scale commercial farms) were subjected to *post-mortem* examinations at two different private veterinary diagnostic laboratories in the northern part of Bangladesh. All these chickens, either dead or sick, were collected from both small- and large-scale commercial farms in five

districts of northern Bangladesh, namely Bogura, Dinajpur, Joypurhat, Rangpur, and Panchagar. Types of chickens included in the *post-mortem* examination were backyard chickens ($n = 10$), Sonali (cross-bred) chickens ($n = 43$), and boiler chickens ($n = 33$). Diagnosis of colibacillosis was made through *post-mortem* examination after confirming the characteristic liver and heart lesions with white and adhering fibrinous materials called perihepatitis and pericarditis, respectively. Heart and liver samples with clear lesions only were subjected to culture immediately for isolation and identification of APEC.

Isolation and identification of *E. coli*

The deep stabbing swabs collected from liver or heart samples were inoculated into Buffered Peptone Water (Oxoid, UK) and mixed thoroughly by vortexing. Then 200 μ L Buffered Peptone Water containing samples was then cultured onto Chromocult[®] Coliform Agar media (Sigma-Aldrich, USA) and incubated at 37°C for 24 hours. Colonies suspected to contain *E. coli* were then further subcultured on MacConkey agar plates and confirmed as *E. coli* by biochemical tests. All isolates were stored in –80°C in a bacterial freeze medium (Luria broth; BD Sparks, USA, phosphate-buffered saline containing 0.45% sodium citrate, 0.1% magnesium sulfate, 1% ammonium sulfate, and 4.4% glycerol) and then later shipped to Sweden for detail phenotypic and molecular analysis. Immediately after arrival in Sweden, all samples were cultured on Uriselect 4 plates (Bio-Rad Laboratories, Hercules, CA, USA) at 37°C overnight to achieve pure cultures. Biochemical tests (ONPG, SIM, VP, OF, and Urease) were also performed for reconfirmation of *E. coli* at the species level.

Antimicrobial susceptibility testing and extraction of genomic DNA

Isolates were cultured on blood agar plates for antibiotic susceptibility testing. Each isolate was subjected to antibiotic susceptibility testing using a panel of antimicrobial agents: cefepime (30 μ g), cefoxitin (30 μ g), cefotaxime (5 μ g), ampicillin (10 μ g), meropenem (10 μ g), gentamicin (10 μ g), amikacin (10 μ g), mecillinam (10 μ g), tigecycline (15 μ g), ciprofloxacin (5 μ g), chloramphenicol (30 μ g), nitrofurantoin (100 μ g), aztreonam (30 μ g) and trimethoprim-sulphamethoxazole (25 μ g). All antibiotic compounds were provided by Oxoid Ltd (Basingstoke, Hampshire, England). Resistance levels to antibiotics were determined by disk diffusion on Mueller–Hinton agar according to the recommendation of the European Committee on Antimicrobial Susceptibility Testing (EUCAST v7.1), and *E. coli* ATCC 25922 was used as a quality control strain. Randomly selected isolates ($n = 28$) showing resistance to one or more of the tested antibiotics were subjected to MIC testing using colistin MIC strips (Liofilchem, Italy) as recommended by EUCAST. *Escherichia coli* ATCC 25922 was used as a quality control strain and *E. coli* NCTC 13846 (*mcr-1* positive) was used as a positive control strain.

Around two to four fresh colonies from blood agar plates were selected for DNA extraction. A Maxwell[®] 16 Cell DNA Purification Kit (Promega, Madison, USA) was used with the automated Maxwell[®] 16 SEV Instrument in order to obtain purified genomic DNA from the bacteria. Once extracted, each genomic DNA was centrifuged for 10 minutes at 16089 g and

the supernatant was taken for storage at -20°C for further molecular analysis.

Detection of antibiotic, biocide, and metal-resistant genes

All isolates were tested for the presence of the colistin resistance *mcr-1* gene by a PCR method¹⁵ recommended by the EU Antimicrobial Resistance Reference laboratory in 2017. Isolates displaying decreased susceptibility (17–20 mm) to cefotaxime were tested for *bla*_{CTX-M} ESBL genes by PCR. The screening of the *bla*_{CTX-M} genotypes was performed using a previously described PCR protocol displaying the designation (CTX-M-I/II/III/IV) of CTX-M positive isolates.¹⁶ Ciprofloxacin-resistant isolates were screened for the presence of several quinolone resistance genes: *qnrA1*, *qnrB1*, *qnrS1*, and *aac(6')-lb-cr* by PCR.¹⁷ Screening of 16S-methylase encoding genes (*rmtA*, *rmtB*, *rmtC*, *rmtD*, *armA*) were performed among aminoglycoside-resistant isolates by simplex and multiplex PCRs.¹⁸ Respective positive controls were used in every PCR reaction. Screening of *merA* gene conferring resistance to the heavy metal “mercury” was performed using the method described earlier¹⁹ and ESBLO610747 strain was used as a positive control for *merA*. PCR assays were performed to investigate the presence of several silver resistance genes (*silE*, *silP*, and *silS*). Cycling parameters were as follows: 95°C for 2 minutes, 40 cycles of 1 minute at 95°C , 1 minute at 55°C , and 3 minutes at 72°C followed by a final extension at 72°C for 5 minutes. The following primers were used: *silE* F(5'-GTACTCCCCCGACATCACTAATT-3') and *silE* R(5'-GGCAGACTGACCGTTATT-3'); *silP* F(5'-CATGACATATCCTGAAGACAGAAAATGC-3') and *silP* R(5'-CGGGCAGACCAGCAATAACAGATA-3'); *silS* F(5'-GGAGATCCCGGATGCATAGCAA-3') and *silS* R(5'-GTTTGCTGCATGACAGGCTAAAGACATC-3').¹² *Klebsiella pneumoniae* strain U-0608239 (carrier of the complete *sil* gene complex) was used as a positive and PCR water was used as a negative control.

Collection of antibiotics use information

During clinical investigations at the veterinary diagnostic labs, farmers were asked for details about the selection and use of antibiotics in their farms. A pretested questionnaire was filled up during sample collection by asking the farmers and recording answers on an MS Excel spreadsheet. The demographic information included information about the poultry farms such as an address, chicken type, and age of chickens and antibiotic use information such as the commercial-generic name of antibiotics, the target of antibiotic use (therapy or growth promoter), duration of antibiotic use, and prescribers of antibiotics.

Results

Antibiotic resistance phenotypes

In total, 55 APEC isolates were isolated from 86 samples. Phenotypic antibiotic resistance profiling confirms only one isolate was completely susceptible, the rest of the 54 isolates displayed resistance ranging from one to six different antibiotics tested. Of these 14 tested antibiotics, resistance was found against 10 antibiotics and the percentage of isolates resistant to antibiotics were: ampicillin (86%), ciprofloxacin (86%), trimethoprim-sulphamethoxazole (73%), chloramphenicol (33%), mecillinam (13%), gentamicin (11%), cefoxitin (11%),

cefotaxime (9%), tigecycline (2%) and nitrofurantoin (2%). None of the isolates were resistant to ceftazidime, meropenem, amikacin, or aztreonam. Multidrug resistance features were found in 35% ($n = 19$) of the isolates, confirming that they were resistant to more than three different antibiotic classes. The most common multiresistance phenotype was CIP-AMP-SXT (ciprofloxacin-ampicillin-trimethoprim/sulphamethoxazole). Details regarding the phenotypic diversity of resistance are listed in Table 1. MIC strip test of 28 selected *E. coli* isolates confirmed none of the isolates showed resistance phenotypically (MIC < 2 mg/L), and MIC was ranging from < 0.016 – 0.125 $\mu\text{g/mL}$.

Detection of antibiotic, biocide, and metal-resistant genes

No isolate was positive for the *mcr-1* gene. Ciprofloxacin-resistant isolates were positive for quinolone resistance genes like *qnrS1* and *aac(6')-lb-cr*; however, none of them were positive for *qnrA1*, *qnrB1*, and *qnrC* genes. The *qnrS1* gene was carried by 21 (38.1%) isolates and the *aac(6')-lb-cr* gene was carried by 8 (14.5%) of the isolates. Only two (3%) isolates were carrying both *qnrS1* and *aac(6')-lb-cr* genes. There was a complete absence of the *bla*_{CTX-M} gene in the collection of cefotaxime-resistant isolates. Screening of 16S-methylase encoding genes (*rmtA*, *rmtB*, *rmtC*, *rmtD*, *armA*) confirmed the complete absence of these genes. The silver resistance marker *silE* was found in four (7.2%) isolates. However, the other tested silver resistance markers such as *silS* and *silP* were not found in the collection of isolates. In total, 21 (38.1%) isolates were positive for the mercury resistance marker *merA*. Of these biocide- and metal-resistant isolates, 10 were carrying either *silE* or *merA* gene without any other ciprofloxacin-resistant genes, but the remaining isolates were carrying *silE* or *merA* gene along with ciprofloxacin-resistant gene. Regarding the coexistence of multiple genes (> 2), one isolate was found to carry *silE*, *merA*, and *aac(6')-lb-cr* genes, and one isolate was found to carry *merA*, *qnrS1*, and *aac(6')-lb-cr* genes. Distribution of Antimicrobial-

TABLE 1. DIVERSITY OF ANTIBIOTIC RESISTANCE PHENOTYPES IN APEC

Phenotypes	Isolate number
CIP	4
MEL	1
FOX	1
CIP-AMP	5
CTX-MEL	1
AMP-C-SXT	5
CIP-AMP-SXT	17
CIP-AMP-MEL	1
CIP-AMP-C-SXT	7
CIP-CN-F-AMP	1
FOX-CIP-AMP-SXT	1
CIP-CN-AMP-C-SXT	2
CTX-CIP-AMP-MEL-SXT	4
FOX-CIP-CN-AMP-C-SXT	3
FOX-TGC-CIP-AMP-C-SXT	1

AMP, ampicillin; C, chloramphenicol; CIP, ciprofloxacin; CN, gentamicin; F, nitrofurantoin; FOX, cefoxitin; MEL, mecillinam; SXT, trimethoprim-sulphamethoxazole.

Metal-, and Biocide-Resistant Genes in APEC Isolates is listed in Table 3.

Use of antimicrobial compounds by farmers

In this study, 10 backyard farmers were not using any kind of antimicrobial compounds for their chickens as they were committed to have “antibiotic free farm,” yet the rest of the 76 (88.4%) farmers were using one to four different antimicrobial compounds in their farms. In total, 60 (78.9%) farmers were using antimicrobial compounds for treatment purposes when there was an outbreak of bacterial and viral infections in the farm, but the other 16 (21%) farmers were using antimicrobial compounds as growth-promoting agents mixed in the water supply. Among these 76 farmers, 33 (43.4%) were using antimicrobial compounds after getting a prescription from veterinarians and 43 (56%) farmers were using different combinations of antimicrobial compounds without any prescription from a veterinarian. Antimicrobial compounds used as growth promoters by the farmers were used without any veterinary consultations. Thirty-nine farmers were using the antimicrobial compounds during the whole course of farming time and 37 farmers were using antimicrobial compounds for a certain period of time depending on the disease condition of the farms. Regarding the use of antimicrobial compounds by these 76 farmers, ciprofloxacin was most extensively used, as at least 41 farms (53.9%) were using this compound. Oxytetracycline was the second most commonly used antibiotic, as around 33 farms (43%) were using this compound. Use of other antimicrobial compounds by farmers was as follows: colistin ($n = 27$; 35.5%), amoxicillin ($n = 24$; 31.5%), doxycycline ($n = 19$; 25%), enrofloxacin ($n = 10$; 13.2%), tylosin ($n = 8$; 10.5%), trimethoprim-sulfamethoxazole ($n = 5$; 6.5%), tiamulin ($n = 3$; 3.9%), ampicillin ($n = 1$; 1.3%), and gentamicin ($n = 1$; 1.3%). These antimicrobial compounds were used either singly or in combination with each other. Detailed information regarding the type and combination of antimicrobial compounds used by farmers is listed in Table 2.

Discussion

APEC infection is a year-round serious problem in the poultry industry of Bangladesh.^{4,14,20} Indiscriminate use of antimicrobial compounds in poultry farms to tackle multidrug-resistant APEC and other bacterial infections has stimulated our interest in exploring the frequency of antimicrobial resistance and their genetic determinants of interest. There is a scarcity of data about antibiotic use and antibiotic resistance among APEC isolates burdening the local poultry production in Bangladesh. This study was conducted focusing on some districts known as the second poultry-production hub in Bangladesh.

Indiscriminate, over-, and misuse of antimicrobial compounds were confirmed in the farms based on the information about antimicrobials used provided by the farmers. Farmers were found to use last-resort antibiotics such as colistin, which is banned for use in livestock. This study also found that critically important human antibiotics like ciprofloxacin, colistin, and gentamicin are extensively used in the Bangladeshi poultry industry with or without prescriptions from veterinarians. More alarmingly, broad-spectrum, critically important human antibiotics such as ciprofloxacin and colistin were used by the farmers as growth-promoting agents without prescription or

TABLE 2. ANTIMICROBIAL COMPOUND USE HISTORY AND PATTERN IN COMMERCIAL POULTRY FARMS OF BANGLADESH

<i>Antimicrobial compounds used</i>	<i>Number of farms</i>
No use of antimicrobial compound	10
Amoxicillin	2
Ciprofloxacin	1
Doxycycline	1
Oxytetracycline	4
Trimethoprim-sulfamethoxazole	2
Tiamulin, doxycycline	3
Tylosin, doxycycline	4
Colistin, trimethoprim-sulfamethoxazole	1
Doxycycline, oxytetracycline	3
Enrofloxacin, ciprofloxacin	5
Enrofloxacin, amoxicillin	2
Ciprofloxacin, oxytetracycline	2
Colistin, amoxicillin	2
Colistin, ciprofloxacin	9
Ciprofloxacin, amoxicillin	8
Amoxicillin, oxytetracycline	3
Colistin, tylosin	1
Trimethoprim-sulfamethoxazole, ciprofloxacin	1
Ciprofloxacin, ampicillin, colistin	1
Ciprofloxacin, amoxicillin, oxytetracycline	1
Tylosin, doxycycline, oxytetracycline	3
Ciprofloxacin, doxycycline, oxytetracycline	1
Ciprofloxacin, enrofloxacin, oxytetracycline	1
Colistin, Ciprofloxacin, Oxytetracycline	1
Colistin, doxycycline, oxytetracycline	3
Oxytetracycline, amoxicillin, colistin	1
Oxytetracycline, ciprofloxacin, colistin	2
Ciprofloxacin, colistin, oxytetracycline, gentamicin	1
Colistin, doxycycline, oxytetracycline, ciprofloxacin	1
Oxytetracycline, ciprofloxacin, amoxicillin, colistin	4
Oxytetracycline, ciprofloxacin, amoxicillin, enrofloxacin	1
Trimethoprim-sulfamethoxazole, ciprofloxacin, enrofloxacin, oxytetracycline	1

recommendations from veterinarians. In Bangladesh, the use of antimicrobial compounds for subtherapeutic purposes (growth promotions) has been prohibited²¹ but still the poultry farmers in Bangladesh were using broad-spectrum antibiotics in their farms. In a previous study from Bangladesh in 2010, 82% (215 of 260 farmers) of the poultry farmers were reported using antibiotics without prescriptions from veterinarians.⁴ In this study conducted in 2016, we found 56% (43 of 76 farmers) of poultry farmers were using various antimicrobial compounds without veterinary consultation. It is because of the Bangladeshi government's recent effort via a massive campaign and training programs for farmers and veterinarians to promote proper use of antimicrobial compounds in the animals and livestock. Another major concern and fear was that the majority of nonprescription-based poultry farmers were using combinations of antimicrobials compounds as cocktail like ciprofloxacin-colistin combined with other antibiotics, which is indeed a matter of great concern. Possibly, low veterinary regulation and low enforcement of the act may be the

TABLE 3. DISTRIBUTION OF ANTIMICROBIAL-, METAL-, AND BIOCIDES-RESISTANT GENES IN APEC ISOLATES IN BANGLADESH

APEC Isolate ID	Antimicrobial-, metal-, and biocide-resistant genes			
	<i>aac</i> -(6')-Ib-cr	<i>qnrS1</i>	<i>silE</i>	<i>merA</i>
CL.15.01	+			
CL.15.02		+		
CL.15.04		+		
CL.15.05		+		
CL.15.06		+		
CL.15.07		+		
CL.15.08		+		
CL.15.09	+			
CL.15.10		+	+	
CL.15.11		+		
CL.15.13	+			+
CL.15.14		+		
CL.15.16		+		
CL.15.19	+			
CL.15.20		+		
CL.15.21		+		
CL.15.30	+			
CL.15.31		+		+
CL.15.32			+	
CL.15.33			+	
CL.15.38	+	+		
CL.15.40	+			+
CL.15.41		+		+
CL.15.42	+		+	+
CL.15.43		+		
CL.15.45		+		+
CL.15.46		+		+
CL.15.47	+			+
CL.15.48				+
CL.15.49				+
CL.15.50				+
CL.15.51				+
CL.15.52				+
CL.15.55				+
CL.15.56		+		+
CL.15.57				+
CL.16.CE1		+		+
CL.16.CE2		+		+
CL.16.CE3	+	+		+
CL.16.CE4		+		+
CL.15.CE5				+

root of this situation in Bangladesh.²² It is very important to use antimicrobial compounds in the poultry farms in order to protect the loss of the poultry farms to different infectious agents and to promote a continuous smooth supply of poultry products to the consumers to meet the demand for protein but still need to consider the importance of antimicrobial resistance from a global perspective and should follow strict veterinary consultation. And we still need to consider how to avoid or stop the use of critically important human antibiotics in livestock practices in Bangladesh.

This investigation showed that multidrug-resistant (MDR) APEC isolates colonized in the infected poultry in Bangladesh. A considerable percentage (35%) of MDR *E. coli* was detected and a majority of the isolates had multiresistance phenotypes, ranging from 2 to 6 different antibiotics tested. In a

previous study, we found 36.6% APEC isolates were MDR;⁴ however, we found an almost similar result in this study where MDR was in 35% of isolates. It is scientifically evident that avian *E. coli* frequently shows multiresistance to more than one antimicrobial drug, which is a global concern to the poultry industry.^{4,23}

APEC isolates were highly resistant to antibiotics like ampicillin (86%), ciprofloxacin (86%), trimethoprim-sulphamethoxazole (73%), and chloramphenicol (33%); resistance has emerged to other broad-spectrum antibiotics like mecillinam (13%), gentamicin (11%), ceftiofur (11%), and cefotaxime (9%). This indicates a reduction of treatment options for APEC in Bangladeshi poultry farms as these are the most common veterinary antibiotics. Considering the practical aspects of the association between antibiotic use and resistance in the farms, we found a stronger association between antibiotic use and resistance in APEC; quinolones and beta-lactams antibiotics are the most commonly used antibiotics in poultry farms, and we found resistance in APEC isolates to these antibiotic classes. Our previous study⁴ on antibiotic resistance in APEC showed lower resistance to ampicillin (25.7%) and ciprofloxacin (12.9%), and we found considerable changes to this situation where both ampicillin and ciprofloxacin resistance was 86%. Interestingly enough, we found APEC isolates were resistant to several human antibiotics that were hardly used in poultry farms in Bangladesh. For example, APEC isolates were resistant, even though lower level, to mecillinam, ceftiofur, cefotaxime, and tigecycline. It is a matter of great concern that these poultry isolates became resistant to some clinically (human) important antibiotics where there was no evidence of the use of these antibiotics in poultry practice. Thus, antibiotic resistance found in avian *E. coli* was found to be widespread and is, therefore, a matter of great concern from both human and veterinary aspects (One Health).

Quinolones have a broad-spectrum activity with a range of bacterial pathogens including *E. coli*, therefore, it has been considered effective in the treatment of colibacillosis. Members of the quinolone antibiotic class, for example, ciprofloxacin, norfloxacin, and ofloxacin are commonly used to treat avian *E. coli* infection.²⁴ In this study, we found extensive use of different quinolones such as ciprofloxacin and enrofloxacin in poultry farms, either with or without prescriptions from veterinarians. As a result, quinolone resistance was found to be common among the collections of APEC and presents a formidable treatment problem. Quinolone-resistant strains carried *qnrS1* and *aac*-(6')-Ib-cr genes, which is an indication of genes possibly causing quinolone resistance in APEC. Rapidly transferable quinolone resistance genes like *qnrS* and *aac*-(6')-Ib-cr are extensively found in *E. coli* from human, poultry, and environmental sources^{24,25} and they could easily confer plasmid-mediated transfer to another quinolone-susceptible *E. coli*.²⁶ One of the interesting aspects of this is that *qnr* confers only low-level resistance, and its main contribution is to facilitate the development of quinolone resistance-determining regions in DNA gyrase and topoisomerase IV.²⁷ The *qnrS1* gene was dominating in APEC isolates and several studies reported previously the presence of this gene in *E. coli* from avian sources in Bangladesh.²⁰ Thus, the *qnrS1* gene is widespread among avian *E. coli* populations in Bangladesh.

Quinolone resistance in *E. coli* has been a serious problem in poultry-production systems in many countries. Countries like Sweden, where quinolone is strictly banned for use in livestock or poultry production, have been suffering from this problem.²⁸ On the other hand, quinolones are extensively used in the poultry-production systems of countries harboring large populations like China and Bangladesh, and poultry production in these countries is suffering because of quinolone resistance.^{20,24} Thus, quinolone resistance in avian *E. coli* has emerged as a serious challenge in the global poultry industry.

Colistin is known to be very effective against APEC infection in poultry, and study reports from Spain and Iran confirmed the high efficacy of colistin use in APEC infections.^{29,30} Resistance to colistin was limited to chromosomal mutations but in recent years, plasmid-mediated colistin resistance has emerged, which could rapidly spread between bacterial species and is considered another emerging global issue.³¹ Plasmid-mediated colistin resistance has already been reported in *E. coli* from human and poultry sources, and epidemiological data confirmed that excessive and misuse of colistin is one of the major forces behind this situation globally.³² For example, indiscriminate use of colistin in poultry production in China, Brazil, and Nepal has confirmed increased colistin resistance of 40% or more.^{8,31} A multitude of investigations has confirmed a notable prevalence of colistin resistance in poultry worldwide compared to other food-producing animals, especially in Asian countries.^{31,33,34} In this study, colistin resistance was not found in the APEC isolates as confirmed by both PCR and MIC testing. However, indiscriminate use of colistin was noted by the poultry farmers, which is alarming and could possibly change the current situation in shocking ways.

APEC isolates of this study were found to carry genetic determinants to heavy metals (*e.g.*, mercury) and biocide compounds (*e.g.*, silver), and these compounds are frequently used in hospitals, farms, and agriculture to ensure medication, hygiene, and biosecurity.^{11,35} Resistance genotypes to mercury and silver have started to become common among *E. coli* from birds and environmental samples in Sweden and India.^{12,36} Biocide substances like silver may also affect antibiotic resistance more directly by selecting for porin deficiency and thereby mediating a cross-resistance to beta-lactams.³⁷ No information is available on how much of these substances are used in farming practices in Bangladesh. The presence of silver and mercury resistance determinants among poultry isolates suggested that anthropogenic activities enhanced the various heavy metal and biocide loads in the farm environment, which could possibly be the reason for the presence of these genes in the study isolates. This is the first study reporting potential metal resistance in APEC isolates in Bangladesh.

Conclusion

Our results reported high rates of antibiotic resistance among APEC isolates recovered from different poultry farms in the northern part of Bangladesh, including those drugs with a broad spectrum of activities and critically important human antibiotics. Evidence from this study also recommends continuous monitoring to track APEC transmissions, antibiotic use, and resistance in poultry farms in Bangladesh using high-resolution molecular and epidemiological tools. This also warrants further detailed investigation by high-resolution typing

methods such as whole genome sequencing to explore detailed genomic epidemiology of APEC in Bangladesh to understand transmission dynamics and control. There is an urgent need to establish nationwide antimicrobial resistance surveillance of pathogens of veterinary importance. It is highly recommended to perform antibiotic susceptibility testing of APEC isolates regularly to a panel of both veterinary and human antibiotics before prescribing any antibiotics to poultry farms in order to ensure proper use and minimization of the economic burden from the antibiotic purchase. More government restrictions on antimicrobial use in poultry farms, by both farmers and veterinarians, should be imposed as soon as possible to prevent and control the spread of antimicrobial-resistant bacteria and their genetic determinants in Bangladesh.

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Authors' Contributions

B.H.: Conceptualization (lead), funding acquisition, investigation, methodology, and data analysis and writing. M.Z.A.: Sample collection, investigation, and methodology and writing. G.R.: Conceptualization (supporting), writing—original draft (supporting), and review and editing.

Disclosure Statement

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